

Logistic Regression

Given dataset containing N training example x_i each with associated class y_i
learning rate $(\alpha) = 0.01$, $y_i = 0$ (Non-thunderstorm), $y_i = 1$ (thunderstorm), $n =$
total number of features. Each sample $X = (x_0, x_1, x_2, \dots, x_n)$

Step 1: Compute the value of Hypothesis

$$\text{Hypothesis} = h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

Where parameter $\theta = \begin{bmatrix} \theta_0 \\ \vdots \\ \theta_n \end{bmatrix}$ and $X = \begin{bmatrix} x_0 \\ \vdots \\ x_n \end{bmatrix}$

The value θ is computed using Gradient descent where the value of all θ_j is updated simultaneously

Repeat

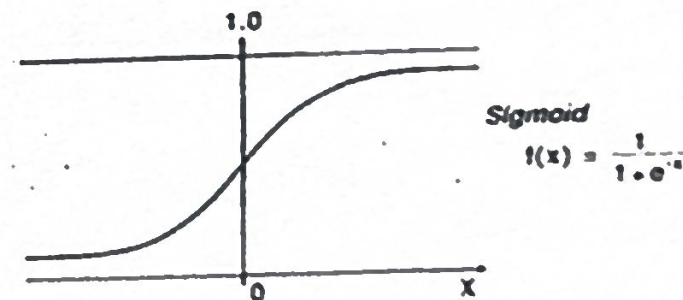
$$\theta_j = \theta_j - \alpha \sum_{i=1}^N (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

Step 2: To make prediction of new X' , put the value of X' in X in the equation

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

If $h_{\theta}(x) \geq 0.5$ C_i is equal to Thunderstorm

If $h_{\theta}(x) < 0.5$ C_i is equal to Non-thunderstorm



Steps of Logistic Regression (LR)

1. Collect the Dataset

Obtain the input features and the target variable. For example, meteorological parameters such as temperature, humidity, pressure, and wind speed can be used for thunderstorm prediction.

2. Preprocess the Data

Handle missing values, treat outliers if required, encode categorical variables, and normalize or standardize features when appropriate.

3. Define Input and Target Variables

Define the input variables as $X = [X_1, X_2, \dots, X_n]$ and the binary target variable as $Y \in \{0, 1\}$. For example, $Y = 1$ represents a thunderstorm and $Y = 0$ represents a non-thunderstorm.

4. Split the Dataset

Divide the dataset into training and testing datasets, such as 70% training and 30% testing, or 80% training and 20% testing.

5. Initialize Model Parameters

Initialize the coefficients $\beta_0, \beta_1, \dots, \beta_n$.

6. Calculate the Linear Combination

Calculate $z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$.

7. Apply the Sigmoid Function

Convert the linear output into a probability using $P(Y=1|X) = 1 / (1 + e^{-z})$. $h_\theta(x)$
 if $h_\theta(x) \geq 0.5$ $ci = \text{class A}$
 if $h_\theta(x) < 0.5$ $ci = \text{class B}$

8. Calculate the Cost Function

Calculate the Binary Cross-Entropy (Log Loss): $J(\beta) = -(1/m) \sum [y_i \log(p_i) + (1-y_i) \log(1-p_i)]$.

9. Optimize the Parameters

Update the model coefficients to minimize the cost function, commonly using Gradient Descent: $\beta_j \leftarrow \beta_j - \alpha (\partial J / \partial \beta_j)$, where α is the learning rate.

10. Repeat the Optimization

Repeat the calculation of the linear combination, sigmoid probability, cost, and parameter update until convergence, the cost becomes sufficiently small, or the maximum number of iterations is reached.

11. Make Predictions

Use the predicted probability $P(Y=1|X)$ and apply a threshold, commonly 0.5: predict 1 if $P(Y=1|X) \geq 0.5$; otherwise predict 0.

12. Evaluate the Model

Evaluate the trained model using Accuracy, Precision, Recall, F1-score, Specificity, ROC-AUC, and the Confusion Matrix.

Overall Flow:

Dataset → Preprocessing → Train/Test Split → Initialize Parameters → Linear Combination → Sigmoid Function → Probability → Cost Function → Parameter Optimization → Prediction → Performance Evaluation

Numerical Example of Logistic Regression Algorithm

This example demonstrates the complete Logistic Regression algorithm for thunderstorm prediction using temperature and relative humidity.

Example Setup

Input features: X_1 = Temperature ($^{\circ}\text{C}$), X_2 = Relative Humidity (%). Target: $Y = 1$ for Thunderstorm and $Y = 0$ for Non-thunderstorm.

Step 1: Collect the Dataset

Example dataset:

Day 1: Temperature = 25°C , Humidity = 60%, $Y = 0$

Day 2: Temperature = 28°C , Humidity = 65%, $Y = 0$

Day 3: Temperature = 30°C , Humidity = 75%, $Y = 1$

Day 4: Temperature = 32°C , Humidity = 80%, $Y = 1$

Day 5: Temperature = 35°C , Humidity = 85%, $Y = 1$

Step 2: Preprocess the Data

Check for missing values, outliers, incorrect values, and categorical variables. For this example, assume the data are already clean.

Step 3: Define Input and Target Variables

$X = [X_1, X_2]$, where X_1 is temperature and X_2 is humidity. $Y = 1$ represents Thunderstorm and $Y = 0$ represents Non-thunderstorm.

Step 4: Split the Dataset

Use, for example, 80% of the observations for training and 20% for testing. Training data are used to learn the model, while testing data are used for evaluation.

Step 5: Initialize Model Parameters

Assume that after training the model obtains:

$$\beta_0 = -12, \beta_1 = 0.20, \beta_2 = 0.10.$$

Step 6: Calculate the Linear Combination

For a new day, suppose $X_1 = 32^{\circ}\text{C}$ and $X_2 = 80\%$.

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

$$z = -12 + (0.20)(32) + (0.10)(80)$$

$$z = -12 + 6.4 + 8 = 2.4.$$

Step 7: Apply the Sigmoid Function

$$P(Y=1|X) = 1 / (1 + e^{(-z)})$$

$$P(Y=1|X) = 1 / (1 + e^{(-2.4)}) \approx 0.917.$$

Therefore, the predicted probability of a thunderstorm is approximately 91.7%.

Step 8: Calculate the Cost Function

Assume the actual class is $Y = 1$.

$$J = -[y \log(p) + (1-y) \log(1-p)]$$

$$J = -\log(0.917) \approx 0.087.$$

The cost is low because the model predicted a high probability for the correct class.

Step 9: Optimize the Parameters

During training, Logistic Regression updates the coefficients to minimize the cost:

$$\beta_j \leftarrow \beta_j - \alpha(\partial J / \partial \beta_j)$$

where α is the learning rate. The gradients determine the direction and magnitude of the parameter updates.

Step 10: Repeat the Optimization

Repeat the sequence Linear Combination $\rightarrow$ Sigmoid $\rightarrow$ Cost $\rightarrow$ Gradient $\rightarrow$ Parameter Update until the model converges, the cost becomes sufficiently small, or the maximum number of iterations is reached.

Step 11: Make the Prediction

The predicted probability is 0.917. Using a threshold of 0.5:

Since $0.917 > 0.5$, the predicted class is $\hat{Y} = 1$.

Therefore, the model predicts: THUNDERSTORM.

Step 12: Evaluate the Model

Suppose the test results are:

$$TP = 40, TN = 45, FP = 10, FN = 5.$$

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN) = 85\%.$$

Precision = $TP / (TP + FP) = 80\%$.

Recall = $TP / (TP + FN) = 88.89\%$.

F1-score = $2 \times (Precision \times Recall) / (Precision + Recall) \approx 84.21\%$.

Complete Logistic Regression Flow

Meteorological Data → Preprocessing → Train/Test Split → Linear Equation → Sigmoid → Probability → Cost Function → Optimization → Prediction → Accuracy, Precision, Recall, F1-score